YEAR 10 SCIENCE
EARTH AND SPACE SCIENCE
TEST 1

NAME: SOLUTIONS

MARK: $\frac{\quad}{60}$

The universe contains features including galaxies, stars and solar systems and the Big Bang theory can be used to explain the origin of the universe

Mark	ND	NW	C	HC	O
Mark	0-20	21-29	30-38	39-44	45-60

QUESTION	ANSWER
1	D
2	D
3	A
4	B
5	B
6	B
7	C
8	A
9	D
10	C
11	D
12	A ?
13	A
14	C
15	D
16	D
17	B
18	B
19	D
20	C
21	C
22	B
23	A
24	C
25	B

1. Patterns of stars in the night sky are called:

   (a) asteroids.
   (b) galaxies.
   (c) nebulae.
   (d) constellations.

2. A light year is defined as:

   (a) the distance light travels from the closest star.
   (b) the distance from Alpha Centauri to the sun.
   (c) the distance of the Earth from the sun.
   (d) the distance light travels in a year.

3. The main fuel of stars is:

   (a) hydrogen.
   (b) helium.
   (c) carbon.
   (d) gold.

4. Clouds of gas and dust that form spectacular images are:

   (a) stars.
   (b) nebulae.
   (c) planets.
   (d) suns.

5. When a star the size of our sun begins to run out of fuel, it transforms and swells in size until it becomes a:

   (a) white dwarf.
   (b) red giant.
   (c) red dwarf.
   (d) white giant.

6. Which one of the following gives the correct order of size from ***smallest to largest***?

(a) Universe, planet, galaxy, solar system.
(b) Planet, solar system, galaxy, universe.
(c) Planet, galaxy, solar system, universe.
(d) Galaxy, universe, planet, solar system.

7. The remains of a supernova forms a:

(a) red giant.
(b) nebulae.
(c) neutron star
(d) white dwarf.

8. A galaxy is a group of many billions of stars held together by:

(a) gravitational forces.
(b) electrical forces.
(c) magnetic forces.
(d) quasars.

9. The Milky Way is most accurately described as a:

(a) planet.
(b) universe.
(c) solar system.
(d) galaxy.

10. The major source of energy in the Sun is:

(a) electrical.
(b) chemical.
(c) nuclear.
(d) magnetic.

11. Many scientists expect that Man will travel to other planets in the solar system within the next 100 years. They do not think, however, that Man will travel to planets of other stars in the universe in the next 100 years.

The main reason that trips to planets of other stars in this period are unlikely is that scientists:

(a) don't know which stars have planets.
(b) fear that the planets may have strange life forms, which could harm the astronauts.
(c) doubt that there are any other stars that have planets.
(d) realise that such trips would take too long because the distances are so great.

12. A star ten times the size of our sun would go through the following stages:

(a) Supernova, neutron star, pulsar, black hole.
(b) Pulsar, neutron star, black hole, supernova.
(c) Neutron star, pulsar, supernova, black hole.
(d) Black hole, supernova, pulsar, neutron star.

13. Most scientists currently favour the theory that the universe:

(a) once fitted inside a pinhead before exploding out into a big bang.
(b) has always existed in its current form.
(c) began on a large scale, and is slowly shrinking, so that it will one day fit inside a pinhead.
(d) has always existed and is continually expanding and contracting.

14. During the nuclear fusion reactions occurring in the Sun's core:

(a) hydrogen particles join to form helium particles, and energy is absorbed.
(b) helium particles join to form hydrogen particles, and energy is released.
(c) hydrogen particles join to form helium particles, and energy is released.
(d) helium particles join to form hydrogen particles, and energy is absorbed.

15. The apparent slow motion of the stars that is observed during the night is due to:

(a) the orbit of stars around other stars.
(b) the orbit of the Earth around the Sun.
(c) the effect of parallax.
(d) the rotation of the Earth.

16. The observed red shift of some stars is due to their:

(a) extremely high temperature.
(b) extremely low temperature.
(c) movement towards the Earth.
(d) movement away from Earth.

17. The 'big bang' theory is a possible explanation of:

(a) the birth of a star.
(b) the expansion of the universe.
(c) the collapse of the universe.
(d) the formation of the solar system.

18. According to the 'big bang' theory, all that existed before the universe began was:

(a) mass.
(b) energy.
(c) space.
(d) time.

19. According to the 'big bang' theory, the universe is:

(a) ripping itself apart.
(b) rotating.
(c) getting warmer.
(d) cooling.

20. The age of the universe is believed by cosmologists to be closest to:

(a) 15 thousand years.
(b) 15 million years.
(c) 15 billion years.
(d) 15 trillion years.

21. The atoms that came together to form the first stars were pulled together by:

(a) electric forces.
(b) magnetic forces.
(c) gravity.
(d) friction.

22. Compared to other stars in the universe, the mass of the sun could be best described as:

(a) huge.
(b) average.
(c) small.
(d) similar.

23. Which of the following is described as a nursery of the stars?

(a) Nebula
(b) Nova
(c) Red giant
(d) White dwarf

24. The colour of a star is most directly related to its:

(a) age.
(b) location.
(c) surface temperature.
(d) volume.

25. Which statement describes evidence that is the basis of the big bang theory?

(a) All galaxies have been measured to be moving at the same speed.
(b) Distant galaxies have been measured to be moving away from the Earth faster than closer galaxies.
(c) All stars have been measured to be moving at the same speed.
(d) Galaxies are expanding as the stars within the galaxy move away from each other.

Short Answer **Write the answer to the questions in the spaces provided.**

1. (a) Name the process occurring in the Sun that produces its energy and briefly explain what is happening. (3)

- Fusion. (1)
- Hydrogen is converted into helium at extremely high temperatures. (1)
- Mass is "lost" during the process and converted into energy according to $E = mc^2$. (1)

(b) The Sun loses 5 million tonnes of mass each second. Explain what is happening to this mass. (2)

- This mass is "lost" during the fusion process. (1)
- It is converted into energy according to $E = mc^2$. (1)

(c) Our Sun is yellow in colour. Why is it this colour and not a different one like red or blue?

(2)

- Colour is determined by surface temperature. (1)
- Our Sun is "average" and has a surface temperature between "cool" (red) and "hot" (blue). (1)

2. (a) All stars begin life from a similar background. Explain how a ***young star*** is formed from the material found in the universe. List the stages involved.

(3)

- Gas and dust collect in a nebula.
- Gravity pulls the material together. (1)
- Pressure increases, causing a temperature increase. (1)
- Fusion starts when temperature $\approx 1 \times 10^6$ °C. (1)

(b) Once formed, stars can follow ***three*** possible paths in their lives. Describe these three pathways.

(3)

- AVERAGE STAR LIKE OUR SUN → becomes a black dwarf. (1)
- LARGE STAR → becomes a neutron star. (1)
- GIANT STAR → becomes a black hole. (1)

(c) Describe the steps involved in a young star ending up forming a ***black hole*** at the end of its life. ***Your steps must be in the correct sequence***.

(4)

- Young star is massive - thousands of times bigger than the Sun. (1)
- Becomes a red supergiant as the hydrogen fuel runs out. (1).
- Outer layers explode off in a supernova. (1)
- Remains > 3 × mass of Sun contract into a black hole. (1)

(d) How is a ***black dwarf*** different from a ***black hole***? Give the characteristics of each.

(4)

BLACK DWARF - no heat. (no fusion)
- cold and dark.
- very small.

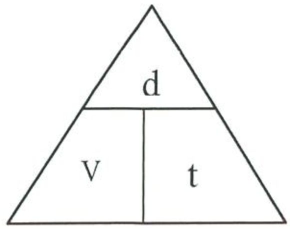
(2)

BLACK HOLE - incredibly dense.
- large gravitational field.
- light cannot escape.

(2).

3. How long does it take light to travel to Earth from the Andromeda galaxy — a distance of 1.4×10^{19} km? (The speed of light can be assumed to be 3.0×10^{5} km/s.)
Set your working out clearly, ***giving your answer in seconds***.

(3)



$v = 3 \cdot 0 \times 10^5$ km/s

$d = 1 \cdot 4 \times 10^{19}$ km

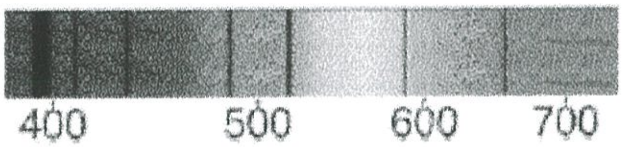
$t = ?$

$t = \frac{d}{v}$ (1)

$= \frac{1 \cdot 4 \times 10^{19}}{3 \cdot 0 \times 10^5}$ (1)

$= \underline{4 \cdot 67 \times 10^{13} \text{ s}}$ (1)

4. The diagram below shows the spectrum from our Sun. The same spectrum is seen from all stars.



(a) Explain briefly what causes the dark lines to form in the spectrum.

(2)

- Cooler gases in the atmosphere of the stars absorbs certain frequencies (colours) of light and re-emit them. (1)
- They appear as dark lines on the rainbow spectrum. (1)

(b) Use the diagram to explain what "red shift" is.

(2)

- For stars and galaxies moving away from Earth, the dark lines are shifted towards the red end. (1)
- This is because the wavelength has increased (or the frequency has decreased) due to the movement. (1)

(c) What does red shift tell astronomers about the universe?

(2)

- Galaxies further away are travelling faster (they are "closer" to the time (1) of the "big expansion").
- The universe is expanding as if it originated from a single point. (1)

5. (a) Give a brief description of how the Big Bang Theory describes the formation of the universe. Include an explanation of how matter came from "nothing".

(3)

- The universe started as a "singularity" - an immensely dense point (1) of energy.
- The singularity expands rapidly to form space and time. (1)
- Pure energy becomes the matter we now see in the universe (1) due to $E=mc^2$.

(b) Edwin Hubble found evidence that supported the Big Bang Theory. Briefly explain what he found and how it supported the theory.

(2)

- Hubble found that the light from distant galaxies had the dark lines on the spectrum shifted significantly towards the (1) red end.
- The further he looked, the greater the "red shift" and the faster the galaxies were moving apart. (1)
- The universe was expanding rapidly from one point.